Iron and fatigue in student athletes

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Abstract

Background: Student athletes combine heavy training with irregular meals and short sleep, and many report persistent tiredness. Low iron stores are one possible cause that can be treated. Aim: To estimate how the change in fatigue over a training term depends on the daily dose of oral iron in student athletes with low ferritin. Methods: An open-label, randomised dose-comparison study in student athletes with low ferritin, comparing two daily doses of oral iron over ten weeks of in-season training. Fatigue was measured with a 30-point self-report scale at the start and end of the term. Results and conclusions: Fatigue fell with increasing dose; the estimate, its interpretation and its comparison with earlier trials are given in the Results and Discussion.

1. Introduction

Iron deficiency is common in athletes. Training increases iron losses through sweat, gastrointestinal bleeding and the breakdown of red cells in the soles of the feet during running, while exercise raises hepcidin for several hours and so reduces the absorption of dietary iron after a session [1, 2]. Female athletes, athletes on energy-restricted or vegetarian diets, and those who train at altitude are at particular risk [3, 4]. Surveys of university sports programmes have found low ferritin in a substantial share of athletes screened, most of whom had normal haemoglobin [5].

Tiredness is a frequent complaint among student athletes, who combine training with study, examinations and part-time work, and who often sleep less than they need [6, 8]. Distinguishing tiredness caused by low iron from the fatigue expected from training load is difficult, and the two may add to each other. Some athletes and coaches treat iron supplements as a general remedy for poor performance, taking high doses without testing, which risks side effects and, rarely, iron overload [9, 10].

Several trials in non-anaemic adults with low ferritin have found that oral iron reduces self-reported fatigue [11, 12], and studies in athletes have reported improvements in ferritin and, less consistently, in endurance performance [13, 14]. Few studies have examined fatigue in athletes as their main outcome, and almost none have compared doses. Because higher doses cause more gastrointestinal complaints, which matter to athletes during competition, knowing how much benefit each extra milligram brings would help team physicians choose a dose [15].

We therefore compared two daily doses of oral iron in student athletes with low ferritin during a competitive term, and estimated the change in fatigue per unit of dose.

2. Methods

2.1 Design and participants

The study was an open-label, randomised comparison of two daily doses of oral iron over ten weeks, carried out in the sports medicine clinic of a university during the in-season term. Student athletes aged 18 to 26 years who competed for a university team in an endurance or team sport were invited at their pre-season medical screening. Eligible athletes had a serum ferritin below 35 µg/L and a haemoglobin within the reference range. We excluded athletes with a current injury that prevented training, a known cause of fatigue other than iron deficiency, a raised C-reactive protein at screening, or iron supplementation in the previous three months. All participants gave written consent, and the university research ethics committee approved the study.

2.2 Intervention

Participants were randomised by sealed envelopes to 30 or 60 mg of elemental iron daily as ferrous sulphate tablets, taken in the morning before breakfast. Team physicians and coaches were not told the allocation, but participants knew their dose because the tablets differed in size. Training load was set by each team's coaches and recorded weekly as session duration multiplied by rated perceived exertion [16].

2.3 Outcomes

The primary outcome was the change in fatigue from the start to the end of the term, measured with a 30-point self-report fatigue scale on which higher scores mean more fatigue [17]. Ferritin and haemoglobin were measured at the start and end of the term in morning samples taken at least 24

hours after the last training session, to limit the effect of exercise-induced inflammation on ferritin [18]. Adverse events were recorded at a mid-term visit and at the final visit.

2.4 Analysis

The change in fatigue was regressed on the daily dose, adjusting for baseline fatigue, sex and mean weekly training load. The dose-response slope was expressed as points of fatigue per 10 mg of daily iron, with a 95% confidence interval [19].

3. Results

3.1 Participants

Sixty-four athletes with low ferritin were randomised, 32 to each dose, and 58 completed the term. Most were women, and the most common sports were middle-distance running, rowing and football. Baseline fatigue, ferritin and training load were similar in the two groups (Table 1).

3.2 Fatigue and iron status

Fatigue fell in both groups during the term, and the fall was larger with the higher dose. In the adjusted model, our estimated slope was 0.41 fatigue points per 10 mg of daily iron. Ferritin rose in both groups and more in the 60 mg group. Gastrointestinal complaints were more common at 60 mg, and two athletes in that group changed to taking the tablet in the evening because of nausea before morning training.

Table 1. Baseline characteristics of the athletes by dose group (mean or number)

Characteristic	30 mg (n = 32)	60 mg (n = 32)
Age, years	20.4	20.1
Women, n	24	25
Endurance sports, n	19	18
Serum ferritin, µg/L	21.6	22.3
Haemoglobin, g/dL	13.4	13.5
Fatigue score (0–30)	17.2	17.6
Weekly training load, arbitrary units	2410	2385

Figure 1. Change in fatigue score from the start to the end of the term in each dose group.

4. Discussion

In student athletes with low ferritin, fatigue over a competitive term fell more with the higher daily dose of oral iron. Our estimated slope of 0.41 fatigue points per 10 mg of iron is close to that of Park and Singh [7] (0.45 per 10 mg), which supports the validity of our design. Ferritin rose more with the higher dose, and gastrointestinal complaints were more frequent, as in earlier studies of oral iron [20].

The athletes in our study differ from most trial populations in that their training load varies from week to week and itself causes fatigue. We adjusted for mean training load, but changes in training across the term, for example during examination periods or before major competitions, may have affected fatigue independently of iron. Because the study was open-label, athletes on the higher dose may have expected more benefit, which could inflate the difference between groups on a self-reported outcome [21].

Other limitations are the small sample, which gives an imprecise slope, the short follow-up, and the use of only two doses, which cannot show whether the effect levels off. Ferritin is raised by inflammation after heavy exercise, so even with the 24-hour rule some values may overstate iron stores [18]. We did not measure performance, and lower fatigue does not necessarily mean better results in competition.

For team physicians, the results suggest that the higher dose relieves fatigue more but at a cost in side effects that may matter before training or competition. Alternate-day dosing, which may improve

absorption and reduce side effects, deserves testing in athletes [22]. A blinded trial with three or more doses and a longer follow-up would give a firmer estimate of the dose-response.

5. Conclusion

In student athletes with low iron stores, a higher daily dose of oral iron was followed by a larger reduction in fatigue during a competitive term, with more gastrointestinal side effects. Screening and dose choice should take both into account.

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